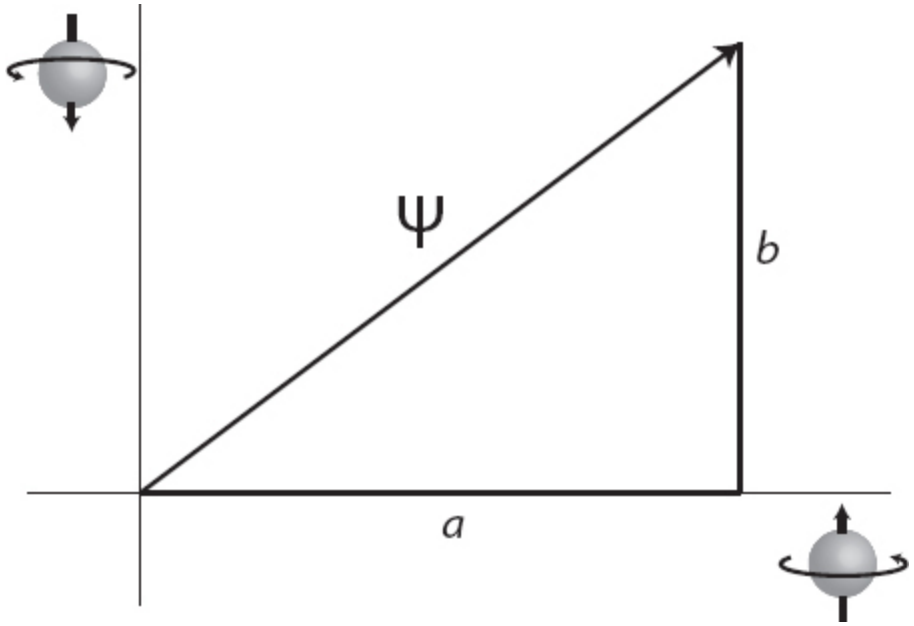
The symbols *a* and *b* are complex numbers, representing the amplitudes for spin-up and spin-down, respectively. The pieces of the wave function representing the different possible measurement outcomes, in this case spin-up/-down, are the "components." In this state, the probability of observing the particle to be spin-up would be $|a|^2$, and the probability for spin-down would be $|b|^2$. If, for example, *a* and *b* were both equal to the square root of 1/2, the probability of observing spin-up or spin-down would be 1/2.

Qubits can help us understand a crucial feature of wave functions: they are like the hypotenuse of a right triangle, for which the shorter sides are the amplitudes for each possible measurement outcome. In other words, the wave function is like a *vector*—an arrow with a length and a direction.

The vector we're talking about doesn't point in a direction in real physical space, like "up" or "north." Rather, it points in a space defined by all possible measurement outcomes. For a single spin qubit, that's either spin-up or spin-down (once we choose some axis along which to measure). When we say "the qubit is in a superposition of spin-up and spin-down," what we really mean is "the vector representing the quantum state has some component in the spin-up direction, and another component in the spin-down direction."



It's natural to think of spin-up and spin-down as pointing in opposite directions. I mean, just look at the arrows. But as quantum states, they are perpendicular to each other: a qubit that is completely spin-up has no